

A Technical Introduction to Overfitting machine learning

Section 1

Benign overfitting Benign overfitting describes the phenomenon of a statistical model that seems to generalize well to unseen data, even when it has been fit perfectly on noisy training data (i.e., obtains perfect predictive accuracy on the training set). The phenomenon is of particular interest in deep neural networks, but is studied from a theoretical perspective in the context of much simpler models, such as linear regression. In particular, it has been shown that overparameterization is essential for benign overfitting in this setting. In other words, the number of directions in parameter space that are unimportant for prediction must significantly exceed the sample size.

Section 2

Regression In regression analysis, overfitting occurs frequently. As an extreme example, if there are p variables in a linear regression with p data points, the fitted line can go exactly through every point. For logistic regression or Cox proportional hazards models, there are a variety of rules of thumb (e.g. 5–9, 10 and 10–15 — the guideline of 10 observations per independent variable is known as the "one in ten rule"). In the process of regression model selection, the mean squared error of the random regression function can be split into random noise, approximation bias, and variance in the estimate of the regression function. The bias–variance tradeoff is often used to overcome overfit models. With a large set of explanatory variables that actually have no relation to the dependent variable being predicted, some variables will in general be falsely found to be statistically significant and the researcher may thus retain them in the model, thereby overfitting the model. This is known as Freedman's paradox.

Section 3

Remedy The optimal function usually needs verification on bigger or completely new datasets. There are, however, methods like minimum spanning tree or life-time of correlation that applies the dependence between correlation coefficients and time-series (window width). Whenever the window width is big enough, the correlation coefficients are stable and don't depend on the window width size anymore. Therefore, a correlation matrix can be created by calculating a coefficient of correlation between investigated variables. This matrix can be represented topologically as a complex network where direct and indirect influences between variables are visualized. Dropout regularisation (random removal of training set data) can also improve robustness and therefore reduce

over-fitting by probabilistically removing inputs to a layer. Pruning is another technique that mitigates overfitting and enhances generalization by identifying a sparse, optimal neural network structure, while simultaneously reducing the computational cost of both training and inference.

Section 4

In statistics, an inference is drawn from a statistical model, which has been selected via some procedure. Burnham & Anderson, in their much-cited text on model selection, argue that to avoid overfitting, one should adhere to the "Principle of Parsimony". The authors also state the following.

Section 5

... an underfitted model would ignore some important replicable (i.e., conceptually replicable in most other samples) structure in the data and thus fail to identify effects that were actually supported by the data. In this case, bias in the parameter estimators is often substantial, and the sampling variance is underestimated, both factors resulting in poor confidence interval coverage. Underfitted models tend to miss important treatment effects in experimental settings.

Section 6

In mathematical modeling, overfitting is the production of an analysis that corresponds too closely or exactly to a particular set of data, and may therefore fail to fit to additional data or predict future observations reliably. An overfitted model is a mathematical model that contains more parameters than can be justified by the data. In the special case of a model that consists of a polynomial function, these parameters represent the degree of a polynomial. The essence of overfitting is to unknowingly extract some of the residual variation (i.e., noise) as if that variation represents the underlying model structure. Underfitting occurs when a mathematical model cannot adequately capture the underlying structure of the data. An under-fitted model is a model that is missing some parameters or terms that would appear in a correctly specified model. Underfitting would occur, for example, when fitting a linear model to nonlinear data. Such a model will tend to have poor predictive performance. The possibility of over-fitting exists when the criterion used for selecting the model is not the same as the criterion used to judge the suitability of a model. For example, a model might be selected by maximizing its performance on some set of training data, yet its suitability might be determined by its ability to perform well on unseen data; overfitting occurs when a model begins to "memorize" training data rather than "learning" to generalize from a trend. As an extreme example, if the number of parameters is the same as or greater than the number of observations, then a model can perfectly predict the training data simply by

memorizing the data in its entirety. (For an illustration, see Figure 2.) Such a model will typically fail severely when making predictions. Overfitting is related to both the complexity of the chosen model and how well it is optimized during training. A function class that is too large, in a suitable sense, relative to the dataset size is likely to overfit. Even when the fitted model does not have an excessive number of parameters, it is to be expected that the fitted relationship will appear to perform less well on a new dataset than on the dataset used for fitting (a phenomenon sometimes known as shrinkage). In particular, the value of the coefficient of determination will shrink relative to the original data. To lessen the chance or amount of overfitting, several techniques are available (e.g., model comparison, cross-validation, regularization, early stopping, pruning, Bayesian priors, or dropout). The basis of some techniques is to either (1) explicitly penalize overly complex models or (2) test the model's ability to generalize by evaluating its performance on a set of data not used for training, which is assumed to approximate the typical unseen data that a model will encounter.

Section 7

Increase the complexity of the model: If the model is too simple, it may be necessary to increase its complexity by adding more features, increasing the number of parameters, or using a more flexible model. However, this should be done carefully to avoid overfitting. **Use a different algorithm:** If the current algorithm is not able to capture the patterns in the data, it may be necessary to try a different one. For example, a neural network may be more effective than a linear regression model for some types of data. **Increase the amount of training data:** If the model is underfitting due to a lack of data, increasing the amount of training data may help. This will allow the model to better capture the underlying patterns in the data. **Regularization:** Regularization is a technique used to prevent overfitting by adding a penalty term to the loss function that discourages large parameter values. It can also be used to prevent underfitting by controlling the complexity of the model. **Ensemble Methods:** Ensemble methods combine multiple models to create a more accurate prediction. This can help reduce underfitting by allowing multiple models to work together to capture the underlying patterns in the data. **Feature engineering:** Feature engineering involves creating new model features from the existing ones that may be more relevant to the problem at hand. This can help improve the accuracy of the model and prevent underfitting.